

Closed-Loop Bayesian Optimization of Multi-Material 3D-Printed Tensegrity-Inspired Energy Absorbers

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Tensegrity-inspired architectures, in which rigid struts are suspended within a continuous flexible network, can exhibit tunable nonlinear mechanical responses and favorable strength-to-weight characteristics, motivating their use in lightweight protective and energy-absorbing structures. Multi-material fused deposition modeling (FDM) can now co-print rigid (PLA) struts and flexible (TPU) elements in a single build, but the resulting design space—spanning strut geometry, tension-element cross-section, connectivity topology, and unit-cell tiling—is too large for efficient trial-and-error experimentation.

This work presents a closed-loop experimental campaign guided by Bayesian optimization (BO) for developing multi-material 3D-printed tensegrity-inspired structures optimized for energy absorption under quasi-static compression and drop-weight impact loading. The key contribution is a design–print–test loop that operates directly on physical measurements without relying on finite-element simulation: a Gaussian-process surrogate model is trained on experimental measurements of peak transmitted force, specific energy absorption (SEA), and compaction efficiency, and an acquisition function recommends the next batch of designs after each test round. The workflow is largely automated—design generation, surrogate updating, and acquisition-function optimization proceed without operator input, while specimen loading and fixture placement remain the primary manual steps.

We define a parameterized family of unit cells using a core-wrapping strategy in which rigid PLA struts are enclosed by continuous TPU skins to improve interfacial interlocking. Design variables include strut diameter and length, tension-element width and thickness, number of struts per unit cell, and tiling pattern. Specimens are fabricated on a consumer-grade multi-material FDM system and tested under quasi-static compression and instrumented drop-weight impact.

We will report the progression of BO-selected designs across iterations, the evolution of the surrogate model as data accumulate, and Pareto-efficient trade-offs between SEA and peak transmitted force within the explored design space. We will also discuss practical lessons for operating a closed-loop physical experiment, including print-failure handling, batch-to-batch variability, and exploration–exploitation balance.